



Rare Occurrence of *Maretia planulata* (Lamarck, 1816) from Sunderban Biosphere Reserve, India

J. S. Yogesh Kumar¹ · Arya Sen¹ · Prabhakar Rai¹ · C. Raghunathan²

Received: 1 September 2025 / Revised: 29 December 2025 / Accepted: 10 March 2026
© The Author(s), under exclusive licence to The National Academy of Sciences, India 2026

Abstract

The Sunderban Biosphere Reserve, one of the world's largest mangrove ecosystems, supports diverse and underexplored marine fauna. The present study documents a rare occurrence of the heart sea urchin *Maretia planulata* (Lamarck, 1816) (Echinoidea: Spatangoida) from the estuarine mangrove ecosystem of the Sunderbans, India. Echinoderms are exclusively marine invertebrates, with Echinoidea being a well-represented class in Indian waters; however, *M. planulata* is the only species of the genus reported from India. Previously, its occurrence was restricted to deeper marine habitats of the Gulf of Mannar. A single specimen (length 30 mm; width 26 mm) was collected during surveys conducted between 2024 and 2025 near Balir Island, Bidhyadhari River, as a trash and identified based on diagnostic morphological features. This record represents the second report of *M. planulata* from India and the first from the Sunderban Biosphere Reserve. Its presence in a mangrove-associated estuarine habitat suggests possible ecological flexibility or distributional shifts, highlighting the need for further benthic biodiversity assessments in this dynamic region.

Keywords Echinoidea · Heart sea urchin · Mangrove ecosystem · Range extension · West Bengal

Introduction

The Sunderbans, a vast delta region formed by the confluence of the Ganges, Brahmaputra and Meghna rivers in the Bay of Bengal, represents one of the largest mangrove ecosystems in the world. These forests are recognized as a biodiversity hotspot due to their unique habitat features, including a complex network of tidal waterways, mudflats and salt tolerant mangrove vegetation, which support a wide range of flora and fauna [1].

Echinoderms are exclusively marine invertebrates, very commonly found in habitats ranging from intertidal to deep sea waters. They are classified into five classes viz., Class Crinoidea (Feather star), Asteroidea (Sea star), Ophiuroidea (Brittle Star), Holothuroidea (Sea Cucumber) and Echinoidea (Sea Urchin and Sand dollar). A total of 743 species

belongs 5 classes, 33 orders, 123 families and 353 genera were reported from Indian water. Among them, class Echinoidea contributes 134 species belonging to 12 orders, 33 families, 78 genera [2].

Maretia planulata commonly called the heart sea urchin, belongs to the family Maretidae and is the only species of this genus reported from India. This species is typically distributed in the tropical and subtropical Indo-Pacific and is capable of burrowing into sediments. Previous reports of the species indicate its occurrence at depths up to 310 m from Tuticorin, Gulf of Mannar [3]. Therefore, its occurrence in the estuarine mangrove ecosystem of the Sunderbans is unusual. Such a finding may reflect occasional dispersal or habitat overlap rather than a confirmed ecological shift. At present, the record should be considered a baseline distributional extension rather than evidence of migration or habitat alteration.

✉ J. S. Yogesh Kumar
yogeshkumarzsi16@gmail.com

¹ Zoological Survey of India, Sunderban Regional Centre, Canning, West Bengal 743329, India

² Zoological Survey of India, M- Block, New Alipore, Kolkata 700053, India

Methodology

Surveys were conducted in the Sunderban Biosphere Reserve between 2024 and 2025, during which a single specimen of *Maretia planulata* was collected using a cast net (mesh sizes

20 mm) near Balir Island, Bidhyadhari River, Sunderban Biosphere Reserve (Lat. 22° 04' 35.0 N; Long. 88° 41' 41.2 E; 50 km from the open sea) on 20th September 2024 (Fig. 1A, C). The specimen was identified based on morphological characteristics [3] and identified specimen deposited in the National Zoological Collections of Zoological Survey of India, Sunderban Regional Centre, Canning, West Bengal (Registration No. SBRC/ZSI/ KN10074).

Results

The taxonomical identification was made based on the morphological features given below.

Phylum : Echinodermata Klein, 1778

Class : Echinoidea Schumacher, 1817

Order : Spatangoida L. Agassiz, 1840

Family : Maretidae Lambert, 1905

Fig. 1 A Map showing collection locality of *Maretia planulata* (Lamarck, 1816); **B** *Astropecten indicus* Döderlein, 1888 collected along with *M. planulata*; **C** Collection site at Bali Island



Genus : *Maretia* Gray, 1855

Materials Examined

One specimen; length 30 mm; width 26 mm; height 13 mm; Balir Island (50 km from Open-sea, depth 21 m), Bidyadhari River, Sunderban Mangrove ecosystem (Lat. 22°04'35.0 N; Long. 88°41'41.2 E); 20.09.2024; Coll. by J.S. Yogesh Kumar.

Description

The specimen is elongated, cordiform (Heart-Shaped) with a slightly anteriorly pointed outline. Specimen length 30 mm, width 26 mm, height 13 mm. The aboral (dorsal) surface is convex, while the oral (ventral) surface is relatively flattened. Aboral surface (Fig. 2A, B) a distinct petaloid pattern is present, composed of five petal-shaped

ambulacra, arranged in a star like form. The anterior petal is slightly shorter and narrower compared to the paired posterior petals. Pores are arranged in paired series along the ambulacral petals. Oral Surface (Fig. 2C, D), the peristome (mouth) is small, eccentric and positioned anteriorly. The peristome is surrounded by a narrow depression bordered with short spines. Labrum elongated, extending posteriorly, partly covering the peristome. Posterior surface (Fig. 2E), the periproct (anal opening) is located sub-terminally on the posterior face, surrounded by a cluster of dense spines. The anal opening is oval, slightly elongated vertically. Apical system (Fig. 2B, F), located anterior to the center of the test with small genital and ocular plates surrounding the madreporite. Genital pores are visible under high magnification. Spination (Fig. 2A–F), the test is covered with dense, slender, and flexible spines, pinkish to pale purple in colour. Spines are longer on the lateral margins and posterior end, directed posteriorly which aids in burrowing. Fine hair-like secondary spines are visible under magnification (Fig. 2F). Coloration in preserved condition, spines appear pale pink to whitish with a faint violent tinge, test surface whitish.

Distribution

Worldwide – Sri Lanka, Indonesia, Philippines, Papua New Guinea, Australia, New Zealand, United Kingdom and western Pacific Islands from Japan (North) to Fiji and New Caledonia (South) and Hawaii (East) [3, 4].

Remarks

M. planulata was distinguished by an extremely shallow frontal notch; aboral primary tubercles extending to the tips of the posterior paired petals; strongly crenulate but not deeply sunken or camellate tubercles; absence of coarse adapical interambulacral tubercles; an oval, T-shaped sub-anal fasciole; elongated primordial plates and labrum reaching the third or fourth ambulacral plates; and comparatively weakly constricted adoral ambulacra, distinguishing it from *Hemipatagus* and *Lovenia* (longer primordial plates of the paired interambulacra do not extend the second ambulacral plate). Present record represents the second report of *M. planulata* from India and the first report from Sunderban Biosphere Reserve, West Bengal Coast. The specimen was collected as a trash along with *Astropecten indicus* Döderlein, 1888 (Fig. 1B). Its occurrence in this estuarine mangrove habitat highlights the ecological importance of the Sunderbans.

Discussion

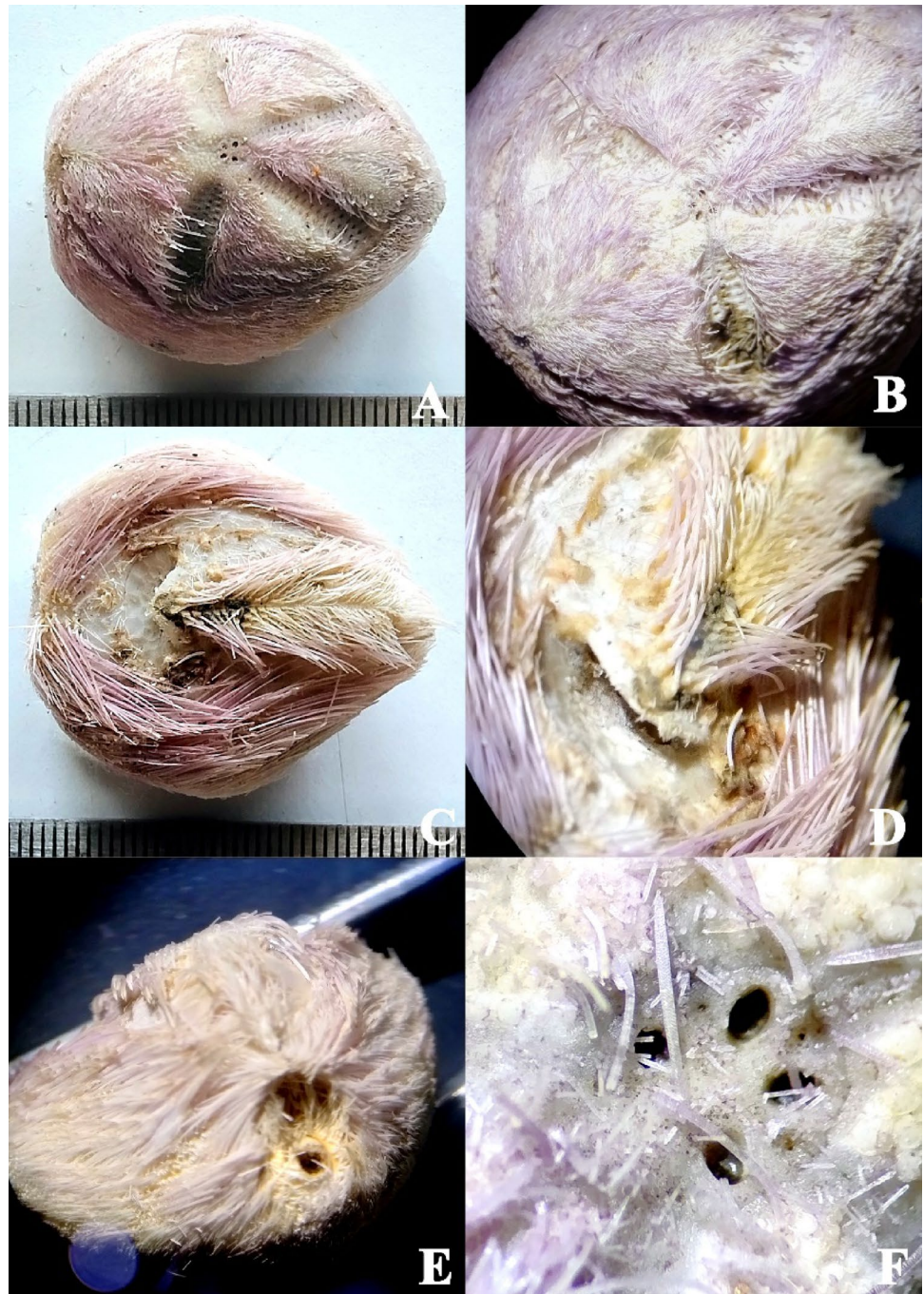
The occurrence of *M. planulata* in the estuarine mangrove environment of the Sunderbans extends its known distribution beyond previously reported offshore habitats such as the Gulf of Mannar. Historically, *Maretia* sp. have been primarily recorded from the Indo-Pacific region, with notable reports from areas such as Sri Lanka, Indonesia, and Australia [5, 6]. The detection of *M. planulata* in the Sunderbans is particularly noteworthy, as previous records from India are limited to the Gulf of Mannar [3]. This finding extends the known distribution range of *M. planulata* into the estuarine mangrove habitats of eastern India, suggesting potential ecological flexibility or biogeographically shifts possibly driven by environmental changes or anthropogenic pressures.

Importantly, an extensive review of Echinoderm fauna reported from India [7–13] indicates that *M. planulata* has not been previously documented from this region. The recent occurrence in the Sunderbans, therefore, represents a new addition to the echinoid fauna of West Bengal and underscores the region's underexplored biodiversity.

Morphologically, the specimen conforms with previous descriptions, exhibiting the characteristic heart-shaped test, petaloid ambulacral pattern, and burrowing adaptations such as posteriorly directed spines [5, 14]. As the present record is based on a single specimen, no definitive conclusions regarding ecological shifts, migration, or anthropogenic influence can be drawn. Single-specimen records are nevertheless important in taxonomy and biogeography, particularly for poorly surveyed habitats such as mangrove-associated benthic systems. No concurrent environmental data were recorded during sampling; therefore, specific ecological drivers cannot be identified at this stage. Also, the dynamic sedimentation, fluctuating salinity, and human interventions like fishing and waste disposal in the Sunderbans could be facilitating this finding, which requires further investigation.

This discovery also highlights the importance of conducting comprehensive and systematic benthic surveys within the region. Given that prior studies [12] have not listed *M. planulata*, this finding suggests that the echinoderm biodiversity of West Bengal, especially within mangrove ecosystems, may be more diverse than currently understood. The presence of *Maretia* could serve as an ecological indicator, reflecting shifts in habitat conditions or faunal distributions associated with environmental stressors.

Fig. 2 *Maretia planulata* (Lamarck, 1816) from Indian Sunderban. **A** and **B** Dorsal view adapical; **C** Ventral view; **D** adoral view with peristome; **E** posterior view with periproct; **F** four gonopores



Conclusion

The present finding underscores the ecological importance of mangrove and estuarine habitats in supporting previously undocumented echinoid taxa and highlights the need for long-term monitoring and integrative taxonomic approaches. The detection of *M. planulata* in this region not only extends its known distribution but also emphasizes the need for comprehensive biodiversity assessments in mangrove and estuarine ecosystems. Future research should

prioritize molecular analyses to elucidate population connectivity and genetic diversity of *Maretia* species in Indian waters. Additionally, long-term ecological monitoring is essential to identify potential reason of any possible long-term shifts driven by climate change, habitat alteration, and human activities, thereby informing conservation strategies for this unique and valuable habitat.

Acknowledgements The authors thank the Director, Zoological Survey of India, Kolkata and Ministry of Environment, Forest and Climate Change, Government of India, for providing facilities and financial

support. Authors also thank the Principal Chief Conservator of Forest (West Bengal), Chief Conservator of Forest and Divisional Forest Officer (Sunderban Tiger Reserve and Sunderban Biosphere Reserve) for their permission and cooperation for this study during the study period 2024–2025.

Author contributions Author JSYK collected the specimen, confirmed the identification and prepared the manuscript. Authors AS, PR assisted in the survey and identification. Author CR verified the identification and assisted in MS preparation.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

References

- Chandra K, Alfred JRB, Mitra B, Chowdhury BR (2017) Fauna of Sundarban Biosphere Reserve, Director. Zoological Survey of India
- Surendar C, Marimuthu N (2024) Fauna of India Checklist: Echinodermata, Version 1.0. Director. Zoological Survey of India
- Vaitheeswaran T, Rajasekaran T, Balasubramani S (2015) Note on the Genus *Maretia* Gray, 1855 and description of *Maretia planulata* (Lamarck, 1861) (Echinoidea, Satangoida), off Thoothukudi Coast of Gulf of Mannar, India (08°35'22.5 N; 78°27'40.9 E) – (310 m). *Int J Mar Sci* 32:1–3
- GBIF (2025) *Maretia planulata* (Lamarck, 1816) in GBIF Secretariat (2023). GBIF Backbone Taxonomy. Checklist dataset <https://doi.org/10.15468/39omei> accessed via GBIF.org on 2025-09-01
- Lawrence JM (1987) A functional biology of Echinoderms. Kent, Croom Helm Ltd. Publishers
- Shaffer FX (1912) Die Echiniden der Mioc anbildung von Eggenburg. *Abhanr Geolog Reichs* 22(2):187–192
- Clark AM, Rowe FW (1971) Monograph of Shallow water Indo-West Pacific Echinoderms. Pitman, London
- Hyman LH (1955) The invertebrates. Echinodermata. McGraw-Hill, New York
- James DB (1989) Echinoderms of Lakshadweep and their Zoogeography. *Bull Cen Mar Fish Res Ins* 43:97–144
- Sastry DRK (2007) Echinodermata of India: an annotated List. *Rec Zool Surv India Occ paper* No 271:1–387.
- Samuel VKD, Krishnan P, Sreeraj CR, Chamundeeswari K, Parthiban C, Sekar V, Patro S, Saravanan R, Abhilash KR, Ramachandran P, Ramesh R (2017) An updated checklist of Echinoderms from Indian waters. *Zootaxa* 4354(1):1–68
- Ramakrishna SJ, Talukdar S (2016) Marine invertebrates of Digha coast and some recommendations on their conservation. *Rec Zool Surv India* 101(3–4):1–23
- Ramakrishna, Jaiswal D, Dinesh KP, Deepak P, Shiva S (2024) Indian Wetlands: A historical perspective. *Rec Zool Surv India* 124(4):411–423
- Mortensen T (1950) A monograph of the Echinoidea Vol 5 Spatangoida . Copenhagen, Brissidae. C. A. Reitzel, pp. 570

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.